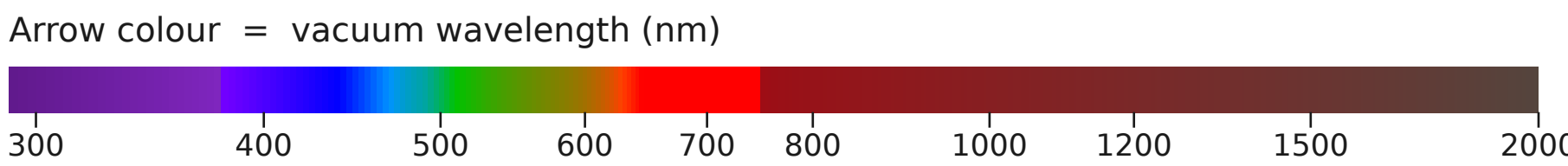


Bi (Bi I). Strongest classified transitions below 2 μm (NIST): 18 levels, 38 lines.

Bi⁺ ionisation limit 58761.65 cm⁻¹ = 7.28552 eV (170.179 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle f || e \mathbf{r} || i \rangle$ in a_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (Ck, M1, E2). Dashed: Rydberg excitation, $n = 70$
 Level caption: NIST energy and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U e J J' \rangle$ (eao)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$4S_{3/2}^o - 2[2]_{3/2}$	195.4710	-	1533.6931	1.138	5.860e+07	-	-	
$4S_{3/2}^o - 2[2]_{3/2}$	206.2286	206.1628	1453.6897	1.581	9.620e+07	-	-	
$4S_{3/2}^o - 2[0]_{1/2}$	211.0886	211.0217	1420.2211	0.916	9.040e+07	-	-	
$4S_{3/2}^o - 2[1]_{3/2}$	222.8905	222.8212	1345.0211	1.387	8.800e+07	-	-	
$4S_{3/2}^o - 2[2]_{3/2}$	223.1304	223.0611	1343.5751	2.775	2.340e+08	-	-	
$4S_{3/2}^o - 2[2]_{3/2}$	227.7262	227.6559	1316.4598	0.742	2.360e+07	-	-	
$2D_{5/2}^o - 5707_{3/2}$	240.1612	240.0881	1248.2968	1.542	5.800e+07	-	-	
$2D_{3/2}^o - 2[2]_{3/2}$	262.8682	262.7899	1140.4666	1.279	4.560e+07	-	-	
$2P_{1/2}^o - 5827_{3/2}$	273.1302	273.0494	1097.6172	0.901	2.020e+07	-	-	
$2D_{3/2}^o - 2[0]_{1/2}$	278.1298	278.0477	1077.8869	0.810	3.090e+07	-	-	
$2D_{3/2}^o - 2[1]_{1/2}$	289.8816	289.7967	1034.1894	1.918	1.530e+08	-	-	
$2D_{5/2}^o - 2[2]_{3/2}$	293.9153	293.8294	1019.9961	2.453	1.200e+08	-	-	
$2D_{3/2}^o - 2[1]_{3/2}$	298.9891	298.9019	1002.6870	1.688	5.400e+07	-	-	
$2D_{5/2}^o - 2[2]_{3/2}$	302.5502	302.4621	990.8851	2.659	8.620e+07	-	-	
$4S_{3/2}^o - 2[0]_{1/2}$	306.8593	306.7702	976.9703	2.182	1.670e+08	-	-	
$2P_{1/2}^o - 2[2]_{3/2}$	359.7123	359.6097	833.4229	1.332	1.930e+07	-	-	

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.